ChartPRM: Process Reward Modeling and Step-Level Preference Optimization for Complex Multimodal Chart Reasoning

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Abstract

Can a compact 3B vision-language model solve multi-step reasoning questions over complex scientific charts? We investigate this question by benchmarking `Qwen2.5-VL-3B-Instruct` on 500 challenging reasoning problems from the CharXiv benchmark. Rather than relying solely on outcome accuracy, we apply dense process-level verification via a multimodal LLM-as-a-judge (`muse-spark-1.1`) to trace where reasoning trajectories diverge. We find that rollouts fail early: over 80% of initial mistakes occur within the first two steps, and an error at step N leads to downstream failure in $\sim$83% of cases. Text analysis of 2,920 judge critiques reveals that failures are overwhelmingly perceptual (axis and legend misreadings comprise 43.5% of errors) rather than logical or arithmetic (1.3%). We then train and evaluate LoRA adapters across five alignment paradigms (SFT, Full DPO, Step-DPO, KTO, and SFT→DPO) on a single consumer GPU. Full-sequence DPO achieves the highest exact-match accuracy at 29% (improving over the 26% base), while SFT enforces 100% formatting compliance at the cost of reasoning accuracy (23%). Although KTO captures the ground-truth answer in text 66% of the time, it suffers from structural formatting collapse. Our findings demonstrate that for small VLMs, the primary bottleneck in chart reasoning is visual grounding rather than chain-of-thought logic.

1 Introduction

2 Related Work

3 The ChartPRM Framework

The central goal of **ChartPRM** is to verify, diagnose, and align step-by-step visual reasoning on complex scientific charts. Scientific plots pack dense information into visual elements. Solving a reasoning question requires multiple distinct steps: locating the relevant panel, reading numerical axis scales, matching curve styles to legend labels, and performing logical comparisons.

ChartPRM decomposes this task into discrete reasoning steps, scores each step individually with a multimodal judge, and uses those step-level signals to guide preference optimization. Crucially, we also ask whether it is possible to align a compact 3B vision-language model using only a very small dataset of verified chart reasoning traces (fewer than 150 examples).

In this section, we describe the four core components of the ChartPRM framework:

1. Structured step-by-step reasoning generation,
2. Multimodal process reward modeling with rollout-level batching,
3. Preference dataset curation and divergence slicing, and
4. Preference alignment objectives and training setup.

3.1 Structured Step-by-Step Reasoning Protocol

To enable step-level verification, the vision-language model must generate reasoning traces that can be parsed into discrete steps. We use a structured Chain-of-Thought prompt that instructs the model to break down its reasoning into sequential steps and end with a clear final answer:

```
Solve the chart reasoning
question step by step.
Structure your answer strictly
as follows:
Step 1:  <visual observation>
Step 2:  <intermediate reading
or deduction>
...
Final Answer:  <concise answer>
```

We enforce three practical formatting rules:

- Each step must start with a explicit label (`Step 1:`, `Step 2:`, etc.).
- The final prediction must appear on its own line after `Final Answer:`, without markdown asterisks.

- The model must not include conversational filler (such as "Certainly! Here is my analysis").

To create a diverse pool of candidate solutions, we sample rollouts using temperature $\tau = 0.7$ and top-$p = 0.9$. For each question in the training pool, we sample up to four candidate rollouts. Across the 500 CharXiv reasoning questions, this procedure generates 1,274 complete rollout trajectories.

We evaluate each rollout with an automated parser that extracts the prediction following `Final Answer:` and applies lowercase whitespace normalization to measure exact match against the ground truth.

3.2 Multimodal Process Reward Modeling

Once rollouts are generated, we evaluate each intermediate reasoning step using a multimodal LLM-as-a-judge. We employ `muse-spark-1.1` (Meta Superintelligence Labs, 2026) as our judge model.

Evaluating each step in an individual API call would require repeatedly uploading the high-resolution chart image for every step, quickly exhausting API token allowances. To address this constraint, we implement *rollout-level batching*: we downscale the chart image to a maximum dimension of 512 pixels (preserving visual legibility while reducing token costs) and evaluate all reasoning steps of a rollout within a single multimodal API request.

The judge receives:

1. The chart image I (resized to ≤ 512 px),
2. The question text q,
3. The reference ground-truth answer y^*, and
4. The full sequence of generated steps $(y_1, \dots, y_T)$.

The judge evaluates each step sequentially in context and returns a structured JSON array containing two fields per step:

- **Binary Step Score ($s_t \in \{0, 1\}$):** The judge assigns $s_t = 1$ if the visual grounding and deduction in Step t are factually correct. It assigns $s_t = 0$ if the step contains an axis misreading, color confusion, hallucinated label, or arithmetic mistake.
- **Natural Language Critique:** For each failed step ($s_t = 0$), the judge provides a concise critique explaining the specific visual or logical failure (for example: "The model misreads the logarithmic y-axis scale as linear, estimating 40 instead of 10").

We define the overall process reward of a complete reasoning trajectory τ as the mean score across all its steps:

$$R(\tau) = \frac{1}{T} \sum_{t=1}^{T} s_t \tag{1}$$

A trajectory is considered *fully verified* only if every step receives a passing score ($R(\tau) = 1.0$). If any step fails, the trajectory receives $R(\tau) < 1.0$.

3.3 Preference Dataset Curation and Step Slicing

From the 1,274 judged rollouts (comprising 4,947 individual steps), we curate training datasets for five distinct alignment methods:

1. Supervised Fine-Tuning (SFT) Dataset: We select only rollouts that achieve a perfect process score ($R(\tau) = 1.0$) and also produce the correct final answer. This strict filter yields 70 gold-standard reasoning rollouts. Because every step is verified by the judge, the model is trained exclusively on factually grounded visual reasoning.

2. Sequence DPO Preference Pairs: For each question where multiple rollouts were sampled, we construct pairwise comparisons (y_w, y_l). The chosen rollout y_w has a higher process reward ($R(y_w) > R(y_l)$) or reaches the correct final answer when y_l does not. This yields 134 chosen-rejected sequence pairs across the training pool.

3. Step-DPO Suffix Slicing: In standard sequence DPO, the entire rejected response is penalized equally. However, our trajectory analysis reveals that rejected rollouts often start with completely correct steps. Penalizing early correct steps teaches the model the wrong lesson.

To solve this credit assignment problem, Step-DPO isolates the first point of divergence. For each pair (y_w, y_l), we compare steps sequentially to identify the first step index t^* where the two traces diverge or where y_l makes its first mistake:

$$t^* = \min\{t \mid y_{w,t} \neq y_{l,t} \text{ or } s_t(y_l) = 0\} \tag{2}$$

We keep the shared prefix $y_{<t^*}$ as prompt context, but compute the preference loss strictly over the divergent suffix starting at t^*. This yields 54 clean, prefix-aligned step pairs.

4. KTO Unpaired Dataset: Unlike DPO, Kahneman-Tversky Optimization does not require paired rollouts for the exact same prompt. We split rollouts into desirable and undesirable pools

Method	Data Size	Epochs	LR	Time
SFT	70 traces	1	1×10^{-5}	12 min
Full DPO	134 pairs	1	1×10^{-5}	22 min
Step-DPO	54 pairs	2	1×10^{-5}	15 min
KTO	84 pos / 252 neg	2	2×10^{-6}	35 min
SFT→DPO	134 pairs	1	2×10^{-6}	24 min

Table 1: **Fine-Tuning Configurations under 16GB GPU Budget.** All models are trained with 4-bit LoRA adaptation ($r = 16$) on `Qwen2.5-VL-3B-Instruct`.

based on their process score and answer accuracy. We collect 84 desirable rollouts ($R \geq 0.75$ and correct final answer) and 252 undesirable rollouts ($R < 0.5$ or wrong answer), establishing a 1:3 positive-to-negative ratio.

3.4 Preference Alignment Objectives and Training Setup

Using the curated datasets, we align the policy model π_θ against a frozen reference policy π_{ref} (the base unaligned instruct model). All models are fine-tuned under a strict 16GB VRAM budget on a single Nvidia Tesla T4 GPU (Google Colab) or an Nvidia Tesla P100 GPU (Kaggle), using 4-bit NF4 quantization (Dettmers et al., 2023) and LoRA adapters (Hu et al., 2022) with rank $r = 16$. Table 1 summarizes the training configurations, dataset sizes, learning rates, and wall-clock runtimes. Peak GPU memory remains under 14.2 GB throughout training.

We investigate five alignment paradigms:

1. Supervised Fine-Tuning (SFT): Following standard alignment pipelines (Ouyang et al., 2022), supervised fine-tuning serves as our initial demonstration baseline by maximizing the log-likelihood of verified gold reasoning steps:

$$\mathcal{L}_{\text{SFT}}(\theta) = -\mathbb{E}_{(x,y_w)} \sum_{t=1}^{|y_w|} \log \pi_\theta(y_{w,t} \mid x, y_{w,<t}) \tag{3}$$

This teaches the model the desired formatting structure, but provides no negative feedback on what errors to avoid.

2. Full-Sequence DPO: Sequence-level Direct Preference Optimization (Rafailov et al., 2023) optimizes a Bradley-Terry preference model directly over token probabilities:

$$\begin{aligned}\mathcal{L}_{\text{DPO}}(\theta; \pi_{\text{ref}}) = -\mathbb{E}\Big[\log \sigma\Big(\beta \log \frac{\pi_\theta(y_w \mid x)}{\pi_{\text{ref}}(y_w \mid x)} \\ -\beta \log \frac{\pi_\theta(y_l \mid x)}{\pi_{\text{ref}}(y_l \mid x)}\Big)\Big]\end{aligned} \tag{4}$$

where σ is the sigmoid function, and $\beta = 0.1$ controls the strength of the KL regularization penalty.

3. Suffix Step-DPO: Suffix Step-DPO (Lai et al., 2024) applies the preference loss only to the divergent suffix:

$$\begin{aligned}\mathcal{L}_{\text{Step-DPO}}(\theta) = -\mathbb{E}\Big[\log \sigma\Big(\beta \sum_{k \geq t^*} \log \tfrac{\pi_\theta(y_{w,k})}{\pi_{\text{ref}}(y_{w,k})} \\ -\beta \sum_{k \geq t^*} \log \tfrac{\pi_\theta(y_{l,k})}{\pi_{\text{ref}}(y_{l,k})}\Big)\Big]\end{aligned} \tag{5}$$

Tokens in the shared prefix $y_{<t^*}$ are masked out from gradient calculation. The loss focuses entirely on correcting the initial visual mistake.

4. Kahneman-Tversky Optimization (KTO): KTO (Ethayarajh et al., 2024) aligns models using unpaired data by applying a prospect-theoretic utility function:

$$\mathcal{L}_{\text{KTO}}(\theta) = \mathbb{E}_y \left[w(y)\left(1 - v_\beta\left(\hat{r}_\theta(x, y) - z_{\text{ref}}\right)\right)\right] \tag{6}$$

where $\hat{r}_\theta(x, y) = \log \frac{\pi_\theta(y|x)}{\pi_{\text{ref}}(y|x)}$ is the implicit reward, $v_\beta(r) = \sigma(\beta r)$ for desirable outputs, $v_\beta(r) = \sigma(-\beta r)$ for undesirable outputs, and z_{ref} is a running estimate of the implicit reward KL baseline.

5. Two-Stage SFT→DPO: We also evaluate a sequential pipeline where the model is first fine-tuned on verified gold traces with SFT for one epoch, and then aligned with sequence DPO using the SFT checkpoint as both the starting policy and the reference policy π_{ref}.

4 DynamicPRM and Test-Time Verification

5 Analysis of Reasoning Steps and Error Taxonomy

6 Alignment Results and Discussion

7 Conclusions

Limitations

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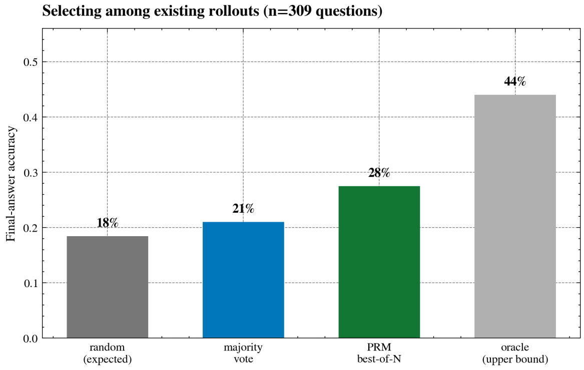

Figure 1: **Inference-Time PRM Verifier Accuracy.** Average step process reward selection (27.5%) outperforms self-consistency majority voting (21.0%) and random rollouts (18.4%) across 309 multi-candidate questions.

Strategy	Holdout EM	Δ vs. Rand	Solvable
Uniform Random	18.4%	–	41.8%
Majority Vote	21.0%	+2.6 pp	47.7%
PRM Best-of-N	**27.5%**	**+9.1 pp**	**62.5%**
Oracle Bound	*44.0%*	*+25.6 pp*	*100.0%*

Table 2: **Test-Time Verifier Performance ($N = 309$ multi-candidate questions).** Selecting candidates by step-level process reward outperforms self-consistency by +6.5 pp.

Trajectory Metric	Count	Share
CharXiv Reasoning Problems	500	100.0%
Evaluated Rollout Traces	1,274	100.0%
Mean Candidates / Question	3.83	–
Total Judged Reasoning Steps	4,947	100.0%
Step Verification:		
– Step Pass Rate (Score = 1)	2,027	41.0%
– Step Failure Rate (Score = 0)	2,920	59.0%
Rollout Outcome:		
– Fully Valid Rollouts (All steps = 1)	154	12.1%
– Rollouts with $\geq$ 1 Step Error	1,120	87.9%
Mean Steps per Rollout	3.88	–
Failure Progression Dynamics:		
– First Error at Step 0	347	31.0%
– First Error at Step 1	546	48.8%
– First Error at Step $\leq$ 1 Cumulative	893	**79.7%**
– Cascade: $P(S_{N+1} = 0 \mid S_N = 0)$	1,630	**82.7%**
– Recovery: $P(S_{N+1} = 1 \mid S_N = 0)$	340	17.3%

Table 3: **Reasoning Trajectory Statistics on CharXiv 500 Pool.** Over 79% of trajectory errors begin within the first two steps. Once an error occurs, downstream steps fail 82.7% of the time.

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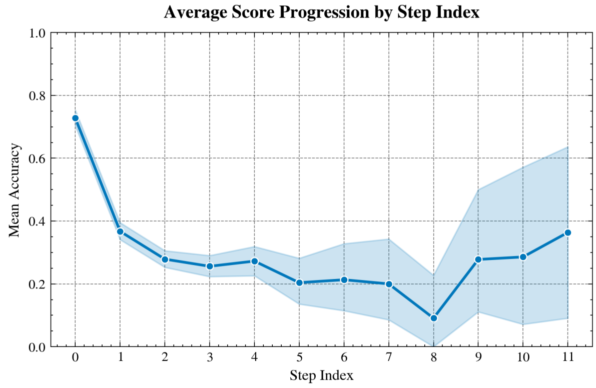
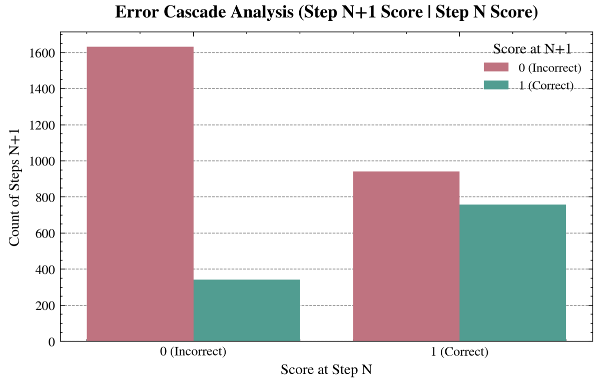
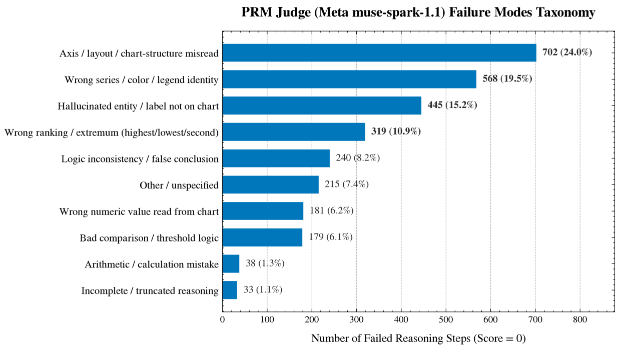


(a) Step Score Cliff by Reasoning Depth

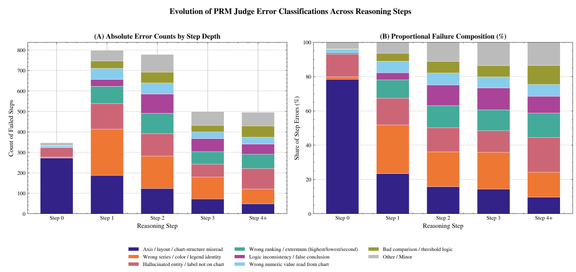


(b) Error Cascade Transition Probabilities

Figure 2: **Reasoning Trajectory Dynamics.** (a) Step accuracy plummets from 73% at Step 0 to 26% at Step 3. (b) When a step fails, the next step also fails 82.7% of the time, proving that early visual mistakes cascade catastrophically.



(a) Distribution of 2,920 Refuted Reasoning Steps



(b) Error Mode Composition Across Step Depth

Figure 3: **PRM Judge Error Taxonomy.** (a) Failures are overwhelmingly perceptual: axis misreads (24.0%) and series confusion (19.5%) account for 43.5% of errors, whereas arithmetic accounts for only 1.3%. (b) Step 0 is 78.5% axis misreads, which triggers downstream ranking and logic errors.

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A Supplementary Training and Diagnostic Visualizations

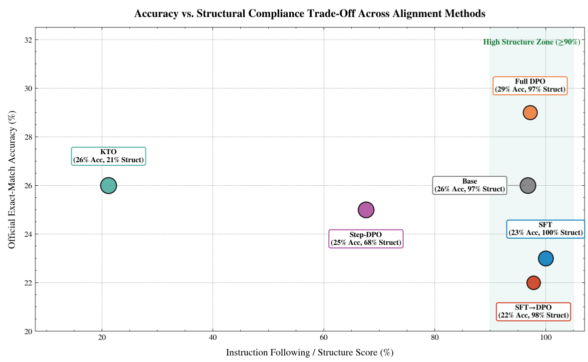

Figure 4: **Accuracy vs. Structure Pareto Frontier.** Bubble diameter is proportional to ground-truth mention recall in prose (47%–66%). Full DPO achieves the best balance (29.0% accuracy, 97.2% structure). SFT attains 100% structure at lower accuracy (23.0%). KTO yields highest latent recall (66.0%) but collapses structurally (21.2%).

Error Category	**Count**	**Share**	**Diagnostic Keywords**	**Representative Judge Critique Excerpt**
Axis / Layout Misread	$N = 702$	24.0%	*axis, subplot, layout, horizontal, vertical*	“Misreads the logarithmic y-axis scale as linear, estimating 40 instead of 10.”
Wrong Series / Color	$N = 568$	19.5%	*blue, line, red, legend, green, curve, orange*	“Attributes the dashed orange curve to Baseline B, but legend associates it with Model A.”
Hallucinated Entity	$N = 445$	15.2%	*hallucinated, absent, not present, visible*	“Refers to a third bar group for year 2020 which does not appear in the figure.”
Wrong Ranking / Extremum	$N = 319$	10.9%	*highest, lowest, largest, maximum, minimum*	“Claims Method 1 achieves highest precision, but Method 3 is clearly higher.”
Logic Inconsistency	$N = 240$	8.2%	*contradicts, concludes, therefore, visual*	“Correctly identifies coordinates 4.2 and 5.1, but falsely concludes 4.2 > 5.1.”
Wrong Numeric Value	$N = 181$	6.2%	*misreads, chart shows, near, false reading*	“Reads point height as 0.65 when the marker aligns with the 0.50 gridline.”
Bad Comparison / Logic	$N = 179$	6.1%	*threshold, exceeds, comparison, logic*	“Fails to evaluate whether the curve remains strictly above the margin.”
Arithmetic Mistake	$N = 38$	1.3%	*miscalculates, arithmetic, computation, sum*	“Correctly extracts values 15 and 8, but computes the difference as 9.”
Incomplete / Truncated	$N = 33$	1.1%	*truncated, incomplete, cuts off, empty*	“Generation terminates prematurely mid-sentence without a verifiable step.”
Other / Unspecified	$N = 215$	7.4%	*unspecified, error, incorrect*	“Vague failure critique without explicit grounding attribution.”
Perceptual Total	$N = 1,270$	**43.5%**	*Combined Axis Misread (24.0%) and Series/Legend Identity (19.5%)*	
Arithmetic Total	$N = 38$	**1.3%**	*Pure calculation failure accounts for only 1 in 77 step errors*	

Table 4: **PRM Judge Error Taxonomy from 2,920 Refuted Reasoning Steps.** Categorized via priority-regex analysis over natural language critiques from `muse-spark-1.1`. Visual grounding (axes, series colors) forms the primary reasoning bottleneck, while arithmetic mistakes are rare.

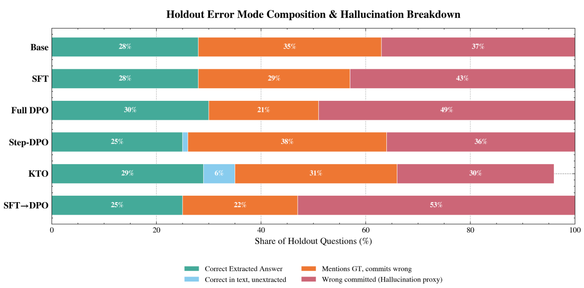

Figure 5: **Holdout Error Mode Breakdown.** Partitioning of test responses into extracted correct, unextracted correct in prose, partial ground-truth mentions, and committed wrong answers (hallucination proxy).

Model	Supervision	EM Acc.	Token	Str+Acc	Recall	Struct	Preamb	Wrong
Qwen2.5-VL-3B	Base Instruct	26.0%	28.0%	26.0%	63.0%	96.8%	7.0%	37.0%
SFT	Gold Traces	23.0%	28.0%	28.0%	57.0%	**100.0%**	**0.0%**	43.0%
Full DPO	Chosen vs. Rej.	**29.0%**	**30.0%**	**29.0%**	51.0%	97.2%	5.0%	49.0%
Step-DPO	Prefix-Masked	25.0%	25.0%	16.0%	64.0%	67.6%	57.0%	36.0%
KTO	Unpaired	26.0%	29.0%	0.0%	**66.0%**	21.2%	99.0%	**30.0%**
SFT→DPO	Two-Stage	22.0%	25.0%	25.0%	47.0%	97.8%	**0.0%**	53.0%
Specialist & Reference-Free Extensions:								
ChartGemma (3B)	Specialist	27.0%	27.0%	–	–	–	–	–
SimPO	Ref.-Free	26.0%	27.0%	26.0%	–	95.0%	6.0%	40.0%
Pareto-DPO	Multi-Filter	30.0%	31.0%	30.0%	–	97.0%	4.0%	46.0%

Table 5: **Main Holdout Benchmark Evaluation on CharXiv ($N = 100$).** EM Acc. is official exact-match accuracy; Token is normalized token match; Str+Acc is format-compliant and correct; Recall measures whether the ground truth is mentioned in text; Struct is structural compliance (0–100%); Preamb is preamble penalty rate; Wrong is wrong committed answers (hallucination proxy).

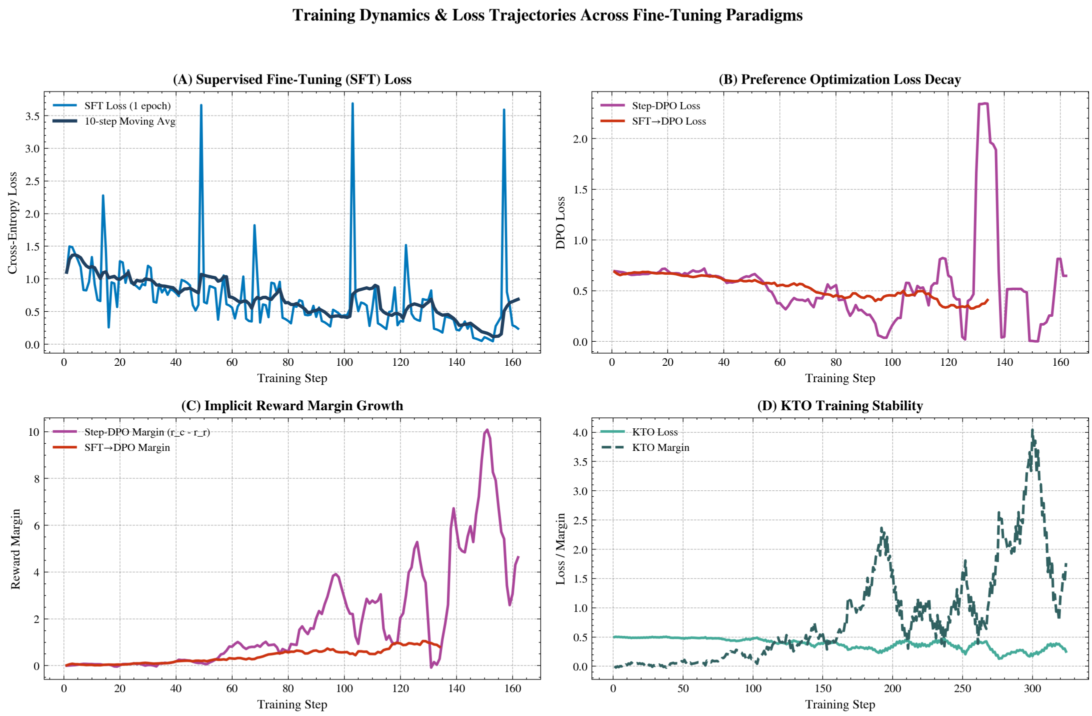

Figure 6: **Training Dynamics Across Alignment Paradigms.** Top-left: SFT cross-entropy loss decay. Top-right: DPO loss trajectories. Bottom-left: Step-DPO implicit reward margin explosion ($\Delta r \rightarrow +10.0$). Bottom-right: KTO margin oscillations under single-T4 GPU constraints.

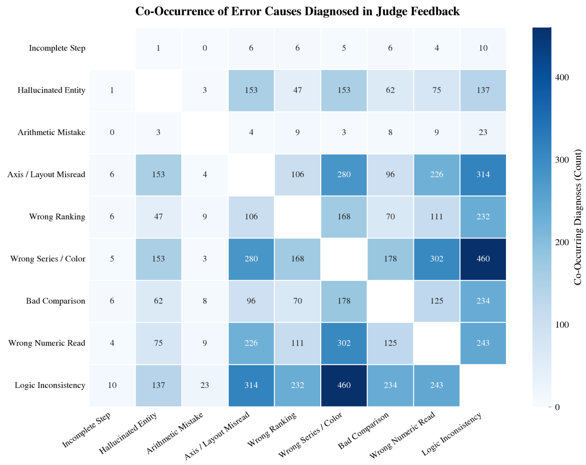

Figure 7: **Compound Error Co-Occurrence Heatmap.** Off-diagonal counts show strong coupling between perceptual errors and downstream logical failures (460 co-occurrences between logic errors and series confusion).

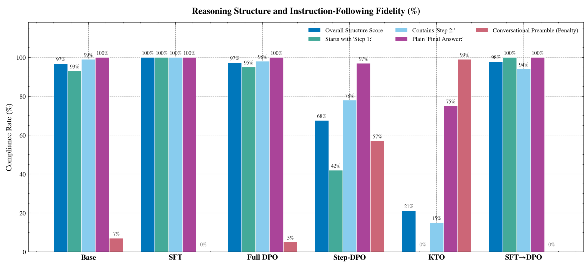

Figure 8: **Structural Instruction Compliance Breakdown.** Evaluates individual prompt constraints. SFT achieves 100% across all tags, while KTO fails primarily due to conversational preambles and missing answer tags.

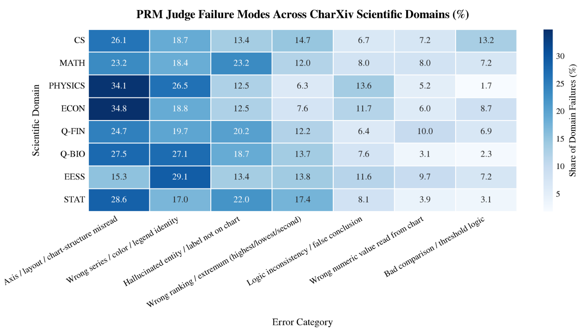

Figure 9: **Error Distribution Across CharXiv Domains.** Physics and Economics charts exhibit the highest rate of axis misreads (>34%), while Mathematics and Computer Science suffer more from entity hallucinations (>22%).